



Conditions for the Effectiveness of Multiple Visual Representations in Enhancing STEM Learning

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Abstract Visual representations play a critical role in enhancing science, technology, engineering, and mathematics (STEM) learning. Educational psychology research shows that adding visual representations to text can enhance students' learning of content knowledge, compared to text-only. But should students learn with a single type of visual representation or with multiple different types of visual representations? This article addresses this question from the perspective of the *representation dilemma*, namely that students often learn content they do not yet understand from representations they do not yet understand. To benefit *from* visual representations, students therefore need *representational competencies*, that is, knowledge *about* how visual representations depict information about the content. This article reviews literature on representational competencies involved in students' learning of content knowledge. Building on this review, this article analyzes how the number of visual representations affects the role these representational competencies play during students' learning of content knowledge. To this end, the article compares two common scenarios: text plus a single type of visual representations (T+SV) and text plus multiple types of visual representations (T+MV). The comparison yields seven hypotheses that describe under which conditions T+MV scenarios are more effective than T+SV scenarios. Finally, the article reviews empirical evidence for each hypothesis and discusses open questions about the representation dilemma.

Keywords Multiple external representations · Visualizations · Conceptual and perceptual knowledge · Sense-making processes · Inductive learning processes

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Introduction

Learning activities vary with respect to how many types of visual representations they use (Fig. 1). For example, consider a student who is learning about fractions. In common learning activities (e.g., tasks in mathematics textbooks), the student may encounter text-only (Fig. 1a) or text plus visual representations (Fig. 1b, c). From prior research on multimedia learning, we know when activities with text plus visual representations are more effective at enhancing students' learning of content knowledge than activities with text-only. For example, visual representations are particularly effective if students have low prior knowledge (expertise reversal effect; Kalyuga and Singh 2015; Mayer and Feldon 2014), if the visual representation adds relevant information (redundancy principle; Schnotz and Bannert 2003), and if students integrate text and visual representations (contiguity principle; Ainsworth 2014; Scheiter et al. 2014; Seufert 2003), among others (see Ainsworth 2008a; Hattie 2012; Mayer 2005, 2009; Schnotz 2005, 2014). However, we know less about how many types of visual representations these activities should include. For example, the fractions student may use text plus a single type of visual representation (T+SV; Fig. 1b) or text plus multiple types of visual representations (T+MV; Fig. 1c). When are T+MV representations more effective than T+SV representations? The goal of this paper is to address this question.

The use of multiple visual representations is common practice in science, technology, engineering, and mathematics (STEM) domains (NRC 2006). Consequently, educational practice guides emphasize the importance of using multiple visual representations (NCTM 2000, 2006; NRC 1996, 2002, 2006), and instruction relies on visual representations in science (Eilam 2013; Kozma et al. 2000; Schönborn and Anderson 2006), technology (Egan and McDonald 2014; Holzinger et al. 2008), engineering (de Jong et al. 2013; Nathan et al. 2011), and mathematics (Arcavi 2003; Noss et al. 1997). According to multimedia learning theory and cognitive learning theories, visual representations can help students learn because they make abstract concepts accessible (Schnotz 2014; Uttal and O'Doherty 2008). Further, multiple visual representations can provide complementary information (Ainsworth 2006, 2014). Consequently, students can construct deeper understanding by integrating them into a

Single external representation

Text-only representation

- A**
- Fractions and sharing**
1. You and eight friends have three pizzas. If everyone eats the same amount of pizza and if you finish all three pizzas, what fraction of a pizza did you eat?
 2. Dan, Joe, Sally, and Sandra have three candy bars, and they share them equally. What fraction of a candy bar does Joe eat? What fraction is eaten by girls?
 3. You run 6 loops. You want to split your run evenly into running at fully speed and running at a leisurely pace. For what fraction of the run will you run at full speed?

Multiple external representations

Text-based plus single visual (T+SV) representations

- B**
- Fractions and sharing**
1. You and eight friends have three pizzas. If everyone eats the same amount of pizza and if you finish all three pizzas, what fraction of a pizza did you eat?
 2. Dan, Joe, Sally, and Sandra have three candy bars, and they share them equally. What fraction of a candy bar does Joe eat? What fraction is eaten by girls?
 3. You run 6 loops. You want to split your run evenly into running at fully speed and running at a leisurely pace. For what fraction of the run will you run at full speed?

Text-based plus multiple visual (T+MV) representations

- C**
- Fractions and sharing**
1. You and eight friends have three pizzas. If everyone eats the same amount of pizza and if you finish all three pizzas, what fraction of a pizza did you eat?
 2. Dan, Joe, Sally, and Sandra have three candy bars, and they share them equally. What fraction of a candy bar does Joe eat? What fraction is eaten by girls?
 3. You run 6 loops. You want to split your run evenly into running at fully speed and running at a leisurely pace. For what fraction of the run will you run at full speed?

Fig. 1 Example scenarios of learning with a single external representation (text-only) (a), or with multiple external representations that contain a text plus single visual (T+SV) representations (b), or text plus multiple visual (T+MV) representations (c)

coherent mental model of the content (Schnotz 2005, 2014; Seufert 2003). In addition, socio-cultural theories suggest that visual representations are an important aspect of learning because they are a major form of communication in scientific and professional communities (Kozma et al. 2000; Wertsch and Kazak 2011).

However, visual representations can confuse students because they play a dual role in education. On the one hand, students have to learn *about* visual representations, that is, how they depict information. On the other hand, students have to learn new information *from* visual representations (Ainsworth 2006; McElhaney et al. 2015). This dual role causes a *representation dilemma*: how can students learn new content from visual representations they do not yet fully understand, and—at the same time—learn how visual representations show content they have not yet learned? For example, in Fig. 1, a student may have to learn about how pie charts depict fractions while learning what a fraction is. Consequently, a major challenge in education is to navigate the representation dilemma, that is, to help students learn *about* visual representations so as to enable their learning *from* visual representations.

Learning *about* representations means that students acquire *representational competencies*: knowledge and skills that enable them to use visual representations to reason and solve tasks (Gilbert 2005, 2008; NRC 2006). Research on multimedia learning has documented that representational competencies enable students to learn *from* visual representations (Ainsworth 2008a; de Jong et al. 1998; McElhaney et al. 2015; NRC 2006). However, the acquisition of representational competencies is costly because it is cognitively demanding and requires instructional support. It stands to reason that these costs increase with the types of visual representations involved. Thus, to understand under which conditions T+MV representations are more effective than T+SV representations, we have to investigate when the benefits of multiple visual representations for students' learning of content knowledge outweigh their costs.

This article approaches this question from the perspective of the representation dilemma in order to extend prior research on multimedia learning, which has mostly focused on *prior* representational competencies that are *prerequisites* to students' learning of content knowledge from visual representations, rather than focusing on the fact that students often acquire representational competencies *while* they are learning content knowledge. To this end, this article integrates theoretical frameworks that describe the processes through which students acquire representational competencies *while* acquiring content knowledge. This focus on learning processes allows to investigate which representational competencies are involved in learning of content knowledge and to analyze how the role these representational competencies play during students' learning of content knowledge is affected by the number of types of visual representations (i.e., T+SV vs. T+MV). This analysis yields seven hypotheses, which describe under which conditions T+MV representations may be more effective than T+SV representations. The article reviews empirical evidence for each hypothesis and concludes by discussing open questions about the representation dilemma.

The following methodological approach was taken in this review article. Research databases were searched for articles published in journals and books, including ERIC, EBSCO, and PsychINFO, and GoogleScholar using the keywords “visualizations,” “graphics,” “visuals,” “diagrams,” “representations,” paired with “learning,” “education,” “instructional design,” or “knowledge.” In addition, published and unpublished proceedings of relevant conferences were reviewed. These conferences included the European Association for Research on Learning and Instruction (EARLI), the American Education Research Association (AERA), the International Conference of the Learning Sciences (ICLS), the International

Conference on Computer Supported Collaborative Learning (CSCL), the Annual Meeting of the Cognitive Science Society (CogSci), the International Conference on Intelligent Tutoring Systems (ITS), the International Conference on Artificial Intelligence in Education (AIED), and the International Conference on Educational Data Mining (EDM). If these conferences had special interest groups (SIGs), their proceedings were also included. Finally, reports from the US National Research Council (NRC) and US educational practice guides on STEM learning were included in this review.

Theoretical Frameworks

Multimedia Learning Theory on Learning with Text and Visual Representations

Multimedia learning theories distinguish several types of representations. Representations stand for something else—the *referent* (Peirce et al. 1935). For example, when we see a picture of a pizza, the referent could be the pizza (a concrete object) or a fraction (an abstract concept). Representations provided in learning activities are defined as *external* representations. When students imagine or mentally manipulate representations, they use *internal representations*. Internal representations are building blocks of *mental models*, which constitute students' content knowledge of a particular topic or domain. The scenario on the left of Fig. 1 uses text-only. *Text* is a *symbolic representation*: a type of external representation that consists of symbols that have arbitrary (or convention-based) mappings to the referent. Symbolic representations can be words in written or spoken language, as well as variables and signs in equations (Schnotz 2014). In contrast, *visual representations* are external representations that consist of icons that have similarity-based mappings to the referent (e.g., diagrams, simulations, or physical models) (Schnotz 2014). Visual representations can differ on a number of design dimensions: they can be concrete (i.e., visual features have high similarity to the referent) or abstract (i.e., visual features have low similarity to the referent) (Braithwaite and Goldstone 2013; Fyfe et al. 2014; Goldstone and Son 2005); they can be physical (i.e., visual features are tangible) or virtual (i.e., visual features are presented on a digital screen) (de Jong et al. 2013); they can be static or animated (i.e., visual features change during the learning experience) (Ainsworth 2008b; Betrancourt 2005); and they can be interactive (i.e., visual features change in response to student interactions) (Moyer et al. 2002; Rasch and Schnotz 2009).

Mayer's (2005, 2009) cognitive theory of multimedia learning (CTML) and Schnotz's (Schnotz 2005, 2014) integrated model of text and picture comprehension (ITPC) draw on information processing theory to distinguish text and visual representations. Text is processed in the verbal part of working memory (even when presented in written form; Mayer and Moreno 2003), whereas visual representations are processed in the visual part (Baddeley 1992, 2012; Chandler and Sweller 1991; Paivio 1986). These different pathways involve three steps that differ between text and visual representations.

First, students *select relevant sensory information* from the external representation for further processing in working memory. To select information from *text*, students engage perceptual processes to analyze the text's surface structure and conceptual processes to analyze the text's semantic structure (Schnotz 2005, 2014). To select information from *visual representations*, students use perceptual processes that capture visuo-spatial patterns of the representation in working memory. To willfully direct visual attention to conceptually relevant

features, students use conceptual processes in top-down thematic selection (Goldstone 1997; Harel 2015).

In the second step, students *organize* this information into an *internal representation* that describes or depicts the information presented in the external representation. For *text*, students form an internal representation that consists of symbolic propositions and describes the surface structure and semantic structure of the text. Drawing on structure mapping theory (Gentner 1983; Gentner and Markman 1997; Gentner et al. 2003), the ITPC proposes that—because the propositions have arbitrary mappings to the referents—the propositions need to be mapped to meaningful referents. This mapping involves perceptual processes that map the propositions to entries in an orthographic lexicon and conceptual processes that map the propositions to prior conceptual knowledge about the domain. The resulting internal representation is descriptive in the sense that it describes the content of the text in the form of symbolic propositions (Schnotz 2005, 2014). For example, the internal representation of the text in task 1 in Fig. 1 may be as follows: SHARE(agents: 9 PEOPLE; amount: 3 PIZZAS). *Visual representations* do not require mapping to an orthographic lexicon because their visual features have similarity-based analog mappings to the referents. Instead, the structure of the visual representation can be directly mapped to the analog internal representations (Gentner and Markman 1997; Gentner et al. 2003; Schnotz 2014). In forming the internal representation, students engage perceptual processes to retrieve perceptual knowledge about the visual appearance of objects based on visual cues. They engage conceptual processes to map the visual cues to conceptual knowledge about these objects. The resulting internal representation is a perceptual analog of the visual representation. It is depictive in that its organization directly corresponds to the visuo-spatial organization of the external visual representation (Schnotz 2005, 2014).

In the third step, students integrate the information contained in the internal representations into a *mental model* of the domain-relevant content knowledge (e.g., schemas, category knowledge). For *text*, students integrate the propositional internal representation into their existing mental model of the content knowledge. They do so by mapping the symbolic propositions to information in long-term memory. For *visual representations*, students integrate the analog internal representation into a mental model by mapping the analog features to information in long-term memory. This third step is what constitutes learning: students learn by integrating the internal representation into a coherent model of the content knowledge (Hegarty and Just 1993; Mayer 2005, 2009; Schnotz 2005, 2014; Wylie and Chi 2014).

In summary, multimedia learning theories suggest that students' learning from visual representations hinges on their ability to form accurate internal representations of the representations' referents and on their ability to integrate internal representations into a coherent mental model of the content—in other words, learning from visual representations requires representational competencies. However, multimedia learning theories do not address the representation dilemma because they do not explain how students acquire representational competencies while also learning content knowledge. The goal of this article is to shed light into this very question.

Cognitive Learning Theories on Learning Processes and Instructional Support

Cognitive learning theories distinguish two broad types of representational competencies: conceptual and perceptual competencies. Although it is well established that conceptual and perceptual competencies are inter-related (Goldstone et al. 1997; Harel 2015), it makes sense to distinguish between them because they are acquired via qualitatively different learning

processes and therefore require different types of instructional support (Goldstone et al. 1997; Kellman and Massey 2013; Koedinger et al. 2012). Cognitive theories align with multimedia learning theory, suggesting that both conceptual and perceptual knowledge play a role during students' learning with visual representations (Mayer 2005, 2009; Schnotz 2005, 2014). They go beyond multimedia learning theory by detailing through which *learning processes* students acquire representational competencies.

Conceptual representational competencies describe knowledge and skills students use to map visual representations to concepts, to make inferences based on visual representations, and to choose a particular visual representation for a task because it depicts relevant concepts. In the CTML and ITPC (Mayer 2005, 2009; Schnotz 2005, 2014), conceptual competencies are involved when students select information for further processing by identifying meaningful visual features, when students form internal representations by mapping visual features to conceptual knowledge, and when they integrate internal representations with prior conceptual knowledge.

Students acquire conceptual competencies via *sense-making processes* (Koedinger et al. 2012). Sense-making processes are verbally mediated because they involve explaining principles of how visual representations depict information (Chi et al. 1989; Gentner 1983; Koedinger et al. 2012). They are explicit because students have to willfully engage in verbal explanations (Chi et al. 1994; diSessa and Sherin 2000). Accordingly, instructional interventions that help students acquire conceptual representational competencies typically engage students in active reasoning, for example by prompting students to self-explain, by having students collaborate on problem-solving tasks or in discussions, or by asking students to draw out their mental models (Fiorella and Mayer 2015; Koedinger et al. 2012; Van Meter and Garner 2005; Wylie and Chi 2014).

Conceptual representational competencies may transfer to novel visual representations because they involve principled knowledge about how visual representations depict content. In particular, structure mapping theory suggests that conceptual competencies may transfer to visual representations with equivalent mappings of visual features to referents (Gentner et al. 2003). For example, students can transfer conceptual knowledge about data visualizations to new types of data representations (Baker et al. 2002).

Perceptual representational competencies involve the ability to effortlessly and efficiently see meaning in visual representations. Gibson (1969) defines perceptual competencies as the improved ability to recognize visual patterns, which results from extensive practice. It is well documented that experts perceive information in representations differently than novices (Chi et al. 1981; Gegenfurtner et al. 2011; Kellman and Massey 2013; NRC 2006). Similar to fluency with a verbal language, perceptual competencies involve fluency in a “visual language,” that is, fluency in processing information shown in visual representations, in using visual representations to solve tasks, and in translating among visual representations (Airey and Linder 2009; Kozma and Russell 2005; Rocke 2010; Wertsch and Kazak 2011). In the CTML and ITPC (Mayer 2005, 2009; Schnotz 2005, 2014), perceptual competencies are involved when students select visual features for further processing, when they form internal representations that depict visuo-spatial patterns, and when they integrate internal representations with prior perceptual knowledge about the appearance of objects.

Students acquire perceptual competencies via *inductive learning processes* (Koedinger et al. 2012). Inductive processes involve both bottom-up processes (perception) and top-down processes (selection) (Goldstone et al. 1997). Inductive processes are automatic processes involved in pattern recognition that are nonverbal in nature (Koedinger et al. 2012;

Richman et al. 1996). Students engage in inductive learning processes when they learn to discriminate, classify, categorize, and become more accurate and efficient in doing so (Koedinger et al. 2012). The expertise literature suggests that this ability to discriminate, classify, and categorize efficiently characterizes perceptual expertise (Goldstone 1997). Inductive processes are considered to be nonverbal because verbal reasoning may not help students acquire perceptual competencies (Kellman and Garrigan 2009; Kellman and Massey 2013) and may even interfere with it (Shanks 2005). Therefore, the acquisition of perceptual competencies does not require direct instruction but involves experienced-based learning through exposure to numerous examples (Gibson 1969, 2000; Kellman and Massey 2013; Richman et al. 1996). Consequently, interventions that help students acquire perceptual competencies expose students to many examples, for example in categorization tasks (Massey et al. 2011; Rau et al. 2015b), through interleaved practice (Rau et al. 2013a, 2014), or in games that involve rapid translations among representations (Eastwood 2013; Moreira 2013).

Research on perceptual learning suggests that perceptual competencies are specific to the type of visual representation. Although perceptual competencies may transfer to novel instances of the same visual representation (e.g., from a pie chart showing $3/8$ to a pie chart showing $7/9$; Kellman et al. 2009; Kellman et al. 2008), they typically do not transfer to different types of visual representations (e.g., from pie chart to numberline) or other domains (e.g., from rectangles showing fractions to bar charts) (Ackerman 2003; Gibson 1969, 2000; NRC 2006).

In summary, cognitive learning theories suggest that the acquisition of conceptual representational competencies involves verbally mediated sense-making processes, and that the acquisition of perceptual representational competencies involves nonverbal inductive learning processes. These different learning processes require different types of instructional supports. However, cognitive theories do not explain how students learning processes unfold over longer learning periods that are typically embedded in social practices in formal learning environments (e.g., classrooms, internships) or informal learning environments (e.g., game play).

Socio-Cultural Theories on Learning with Visual Representations

According to socio-cultural theories, the acquisition of representational competencies enables students to *participate in discipline discourse* and hence to become *legitimate members of scientific and professional communities*. Visual representations are part of discipline discourse in the sense that they are thinking tools for individuals and groups, communication tools, as well as problem-solving tools (Airey and Linder 2009; Gilbert 2008; Kozma and Russell 2005; Wertsch 1997). These tools evolved as part of the cultural history of scientific and professional communities and are therefore an integral part of their cultural practices (Cobb 1995; Vygotsky 1978b). For example, an ethnographic by (Kozma et al. 2000) shows how chemists use visual representations to reason and communicate with other chemists. The use of visual representations serves both cognitive functions (e.g., visual representations allow externalizing one's knowledge) and social aspects (e.g., visual representations facilitate communication and co-construction of knowledge). Accordingly, socio-cultural theories suggest that the goal of classroom-based instruction is to enculturate students in the "ways of thinking" that are common in professional or scientific communities (Lave and Wenger 1991; Reiser and Tabak 2014; Sawyer 2006), which involves enculturation into the use of visual representations (Gilbert 2008).

According to socio-cultural theories, learning of representational competencies first manifests in social interactions, which then become internalized (Vygotsky 1978a; Wertsch 1997; Wertsch and Kazak 2011). These social interactions occur when students participate in discipline discourse that involves both verbal and nonverbal communication (Airey and Linder 2009; Wertsch and Kazak 2011). Through social interactions with a more knowledgeable person (e.g., teacher, another student, parent), students learn how to use visual representations in authentic tasks. During this process, students often start by observing experts and may then use visual representations themselves with the support of the more knowledgeable person. With increasing experience, this support is faded out and students take on increasing responsibility for solving authentic tasks themselves (Collins and Kapur 2014; Collins et al. 1989; Lave and Wenger 1991; Reiser and Tabak 2014). Thus, participation in community practices is a critical socially mediated mechanism through which students acquire representational competencies.

These social interactions mediate students' engagement in both sense-making processes and inductive learning processes. Students engage in verbal sense-making processes when they participate in discourse with other students, with teachers, or with experts. For example, the literature on cognitive apprenticeship describes how students learn about visual representations by observing and appropriating how experts use visual representations to reason about concepts, by collaboratively constructing knowledge about how visual representations depict concepts, and via scaffolded reflections on the representations (Collins and Kapur 2014; Collins et al. 1989; White and Pea 2011). The literature on inquiry learning provides further examples of students' increasingly sophisticated use of visual representations as they learn to make data-driven arguments (Cobb and McClain 2006; Linn et al. 2015).

Further, social interactions engage inductive learning processes that are nonverbal and implicit. Students induce which visual features to attend to by observing experts and by encountering a variety of visual representations over time, without necessarily explicitly reflecting on visual representations. Participating in community practices results in the acquisition of fluency in a "visual language" (Airey and Linder 2009; Kozma and Russell 2005). For example, research on medical education shows that expert surgeons help student trainees acquire perceptual competencies by providing nonverbal cues such as gaze direction and gesturing (Cope et al. 2015). Airey and Linder (2009) provide further examples of physics students becoming fluent in using visual representations by imitating how visual representations are used, without necessarily conceptually understanding meaning at first.

In contrast to cognitive learning theories and socio-cultural theories suggest that representational competencies are tied to the specific cultural context in which it was acquired (Greeno and Engeström 2014; Lave and Wenger 1991). That is, representational competencies may transfer to visual representations that involve the same types of disciplinary discourse (e.g., contentions, use practices). However, representational competencies may not transfer to activities that are part of different community practices or involve different types of disciplinary discourse (Carraher and Schliemann 2002)—even if these activities use the same type of visual representation (e.g., a pie chart of fractions in math class versus a pie chart of poll results in the news).

In sum, socio-cultural theories align with multimedia learning theory and cognitive learning theories in proposing that both conceptual and perceptual representational competencies are key to students' learning in STEM. Socio-cultural theories extend these theories by describing how verbal sense-making processes and nonverbal learning processes unfold over time as students participate in authentic community practices and become increasingly legitimate

members of scientific and professional communities. In particular, socio-cultural theories posit that sense-making processes and inductive learning processes are socially mediated.

Representational Competencies Involved in Learning of Content Knowledge

This article combines these theoretical perspectives to gain insights into the representation dilemma. The following review describes which representational competencies are involved in students' learning content knowledge from visual representations. Specifically, research on conceptual and perceptual representational competencies, as well as meta-representational competencies is reviewed. Table 1 provides an overview of these representational competencies.

Conceptual Representational Competencies

A biology professor reflects on the use of visual representations:

“We biologists use a straight line to represent many things (nucleic acid sequences, protein amino acid sequences, chemical bonds, cell membranes, etc.). Students see these straight lines in textbooks, slideshows, and on the blackboard. We assume our students interpret these lines the same way we do; but is that always true?”

A student asked about challenges with visual representations in chemistry said: “It is confusing because some of the visual representations seem to contradict each other.”

These quotes illustrate the importance of conceptual representational competencies for students' learning of content knowledge. Building on Ainsworth's (1999, 2006, 2014) Design, Functions, and Tasks (DeFT) framework for learning with representations, two types of conceptual competencies are distinguished: *visual understanding* of the individual visual representations and *connectional understanding* of relations among multiple visual representations.

Visual Understanding: Connecting Visual Representations to Concepts

According to multimedia learning theories, students' learning from visual representations requires that they have *visual understanding* of how a visual representation depicts information about domain-relevant concepts (Ainsworth 2006, 2014; Mayer 2005, 2009; Schnotz 2005, 2014). According to the CTML and ITCP, visual understanding involves the ability to map visual representations to prior knowledge about the referent (i.e., to particular aspects of a domain-relevant concept they depict). These connections involve mapping an analog internal representation to conceptual knowledge. Structure mapping theory (Gentner and Markman 1997; Gentner et al. 2003) suggests that students are more likely to establish mappings between visual representations and the referent if the visual features of the representation are similar to the referent. However, these mappings may not necessarily be correct or relevant. If the shared visual features between the representation and the referent are conceptually relevant, these mappings are more likely to be correct; but if the shared visual features are irrelevant, these mappings are more likely to be incorrect. In line with this notion, DeLoache's (2000) dual representation theory suggests that the more similar a visual representation is to its

Table 1 Overview of representational competencies, specific knowledge and skills they involve, and the learning processes through which students acquire them

Representational competencies	Involved knowledge and skills	Learning processes
Conceptual competencies		
Visual understanding	–Ability to connect visual representation to concepts –Ability to identify relevant visual features –Ability to connect visual representations to text –Understanding of general principles about how the visual representation depicts information –Ability to use visual representation to communicate with others –Understanding of conventions for use of visual representation	Verbally mediated sense-making processes that are involved in using a particular visual representation in authentic tasks within the target community
Connectional understanding	–Ability to connect multiple visual representations to one another –Ability to identify relevant similarities among visual representations –Ability to explain and communicate about differences and correspondences between visual representations –Understanding of conventions for using multiple visual representations together	Verbally mediated sense-making processes involved in using multiple visual representations together in social interactions in the target community
Perceptual competencies		
Visual fluency	–Efficiency in connecting visual representations to concepts –Access to direct one-on-one mappings of perceptual chunks and concepts –Ability to think, learn, and express oneself with a visual representation –Ability to infer community-specific knowledge beyond what the visual representation explicitly shows	Nonverbal inductive learning processes involved in using visual representations to participate in disciplinary discourse within the target community
Connectional fluency	–Efficiency in connecting multiple visual representations to one another –Access to direct one-on-one mappings between multiple perceptual chunks –Flexibility to switch between multiple visual representations –Ability to infer community-specific knowledge across multiple visual representations beyond what is explicitly shown	Nonverbal inductive learning processes involved in using multiple visual representations together in disciplinary discourse within the target community
Meta-representational competencies	–Ability to choose visual representation based on task demands –Ability to choose visual representation based on task context –Ability to choose visual representation based on own ability level –Ability to choose visual representation based on personal goals	Verbally mediated sense-making processes and nonverbal inductive learning processes involved in choosing visual representations

referent, the more difficult it is for students to discern which of the similarities are conceptually relevant. Hence, it may be more difficult to form accurate internal representations of visual representations than of text, and it may be more difficult to form accurate internal representations of concrete than of abstract visual representations. Therefore, visual understanding involves the ability to distinguish between conceptually relevant and irrelevant similarities between the representation's visual features and the referent.

In multimedia learning activities, relevant concepts are often described in written or spoken words that accompany a visual representation. Hence, the ITCP and the CTLM propose that visual understanding involves the ability to make connections between text and the visual representation. This involves mapping the analog internal representation (which results from processing the visual representation) to a propositional internal representation (which results from processing words). Eye-tracking research shows that text typically guides the student's inspection of visual representations (Hegarty and Just 1993; Mason et al. 2013a, b). That is, the propositional internal representation guide students' mapping of corresponding features in the internal analog representation. Thus, the ability to make text-visual connections is an important aspect of visual understanding.

Cognitive learning theories suggest that the acquisition of visual understanding involves learning about general principles of how a visual representation depicts information about the referent (DeLoache 2000; Eilam and Ben-Peretz 2012; Uttal and O'Doherty 2008). These principles describe which visual features are conceptually irrelevant, which visual features are meaningful, and how these mappings apply to multiple situations (Gentner et al. 2003). For example, a student learning about the pie charts in Fig. 1 may learn the general principle that pie charts depict fractions as the proportion of shaded area relative to the whole area, which can be quantified as the number of equally sized shaded sections relative to the total number of sections. An irrelevant feature is therefore the color of the sections. A relevant feature is that the sections are equally sized. The student may then apply this principle to pie charts with different colors.

Socio-cultural theories suggest that the acquisition of visual understanding is socially mediated because visual understanding involves the ability to use visual representations to communicate about concepts (Airey and Linder 2009; Kozma and Russell 2005; Wertsch and Kazak 2011). Socio-cultural theories consider visual representations to be a cultural tool that is shaped by the history of a given scientific or professional community (Vygotsky 1978a, b). Therefore, the acquisition of visual understanding involves learning about conventions for the use of the visual representation in discipline discourse (Airey and Linder 2009). For example, visual understanding involves knowing which arguments a visual representation can support (Cobb and McClain 2006) and how to use it in collaboration (Roschelle 1992; Wertsch and Kazak 2011).

There is extensive evidence that visual understanding is an important prerequisite for students' learning of content knowledge (for extensive overviews, see Ainsworth 2006; McNeil and Jarvin 2007; Uttal and O'Doherty 2008). Yet, students' difficulties in understanding visual representations are also well documented (Ainsworth 2006; McNeil and Jarvin 2007; NRC 2006; Uttal and O'Doherty 2008). For example, DeLoache (2000) found that young children have difficulties understanding that visual representations stand for something else because they treat the representations themselves as objects (also see DeLoache and Marzolf 1992). These difficulties are not limited to young children (Tversky 2011). For example, elementary-school students have difficulties mapping visual representations to

concepts (McNeil et al. 2009). Even graduate students often fail to understand how common visual representations depict key concepts (Strickland et al. 2010).

Taken together, multimedia learning theories, cognitive learning theories, and socio-cultural theories suggest that learning *from* visual representations involves visual understanding. Students acquire visual understanding by engaging in sense-making processes that are involved in using a particular visual representation in authentic tasks within the target community.

Connectional Understanding: Connections Among Visual Representations

According to multimedia learning theory, students' learning from multiple representations requires that students integrate the information from them into a coherent mental model. This implies that when students encounter multiple visual representations, they need to understand the connections among the visual representations (Ainsworth 2006, 2014). Making visual-visual representation connections involves different of types of working memory processes than the text-visual connections just discussed with respect to visual understanding. The ITCP (Schnotz 2005, 2014) suggests that students make visual-visual connections by mapping analog internal representations to one another. Recall that visual representations use features that have some level of similarity to the referent, and that analog internal representations maintain these similarity-based mappings. Hence, students make connections among analog internal representations that may have similar mappings to the referent. However, not all of these similarities are conceptually relevant. Some of them are incidental similarities (i.e., surface similarities, such as the fact that both the pie chart and the rectangle in Fig. 1c are gray). These similarities may make it easier for students to make connections (DeLoache 2000; Gentner and Markman 1997). However, if these similarities are not conceptually relevant but incidental, these connections may not be correct. Therefore, connectional understanding involves the ability to distinguish surface-level similarities from conceptually relevant similarities among visual representations.

According to cognitive learning theories, the acquisition of connectional understanding involves sense-making of how one representation can constrain the interpretation of the second representation, and how they complement one another in depicting information (Ainsworth 2006, 2014). That is, students learn to explain correspondences between visual representations (i.e., how both show similar information about domain-relevant concepts), and differences between them (i.e., how they show different information about the concepts) (Rau et al. 2015b). Hence, connectional understanding involves ability to compare visual representations.

According to socio-cultural theories, the acquisition of connectional understanding is socially mediated. Experts tend to use multiple visual representations in conjunction to solve tasks and to communicate with one another (Airey and Linder 2009; Kozma and Russell 2005; Kozma et al. 2000; Schönborn and Anderson 2006). These social practices involve the ability to discuss correspondences and differences between visual representations. In addition, acquisition of connectional understanding involves learning about historical or cultural conventions for using multiple visual representations together to solve specific tasks (Kozma and Russell 2005). This knowledge may also involve an understanding of historical discoveries that resulted in practices of combining multiple visual representations. For example, one visual representation may have been found to be inaccurate for certain applications and is therefore used in conjunction with another visual representation that was developed to address these shortcomings (Rocke 2010).

There is abundant evidence for the importance of connectional understanding for students' learning of content knowledge in the educational psychology literature (Ainsworth 2006) as well as in STEM education literatures (Miura and Yamagishi 2002; Patel and Dexter 2014; Rathmell and Leutzinger 1991; Savec et al. 2009). Consequently, connectional understanding is an important goal in STEM education (NCTM 2000, 2006; NMAP 2008; NSES 2013; NGSS 2013).

In sum, multimedia learning theories, cognitive learning theories, and socio-cultural theories suggest learning *from* multiple visual representations involves connectional understanding. Connectional understanding is acquired via sense-making processes that students engage in when using multiple visual representations together in social interactions in the target community.

Perceptual Representational Competencies

A student reflecting on the role of visual representation for her learning said: “they help tremendously with forming a picture of what actually is happening, but they can be difficult to work with, and they are time consuming.” Conceptual representational competencies do not seem to be the issue for this student, but yet, she seems to struggle when learning with visual representations—potentially because she lacks perceptual representational competencies.

Two types of perceptual competencies are distinguished: *visual fluency* with an individual visual representation and *connectional fluency* in translating among visual representations.

Visual Fluency: Seeing Meaning in an Individual Visual Representation

There is abundant evidence that domain experts “see at a glance” what a given representation shows without any perceived mental effort (Chi et al. 1981; Gegenfurtner et al. 2011; Kellman and Massey 2013; NRC 2006). These experts have *visual fluency*: they are highly efficient in forming accurate internal representations of a visual representation. According to multimedia learning theories, visual fluency involves efficiency in connecting visual representation to concepts (Mayer 2005, 2009; Schnotz 2005, 2014). Because concepts are often described in written or spoken words, visual fluency involves efficiency in mapping propositional to analog internal representations. Indeed, research on expertise shows that experts can quickly determine whether a visual representation shows the same information as text-based representation (Chase and Simon 1973; Kellman and Garrigan 2009; Kellman and Massey 2013).

According to cognitive learning theories, the acquisition of visual fluency involves inductive processes involved in perceptual pattern learning (Gibson 1969; Goldstone and Barsalou 1998; Kellman and Massey 2013; Richman et al. 1996). High efficiency in recognizing perceptual patterns results from perceptual chunking: relevant visual features serve to retrieve a corresponding schema from long-term memory that constitutes the relevant concepts (Richman et al. 1996; Taber 2013). Thus, the acquisition of visual fluency means that students learn direct one-on-one mappings between perceptual chunks and concepts (Koedinger et al. 2012).

According to socio-cultural theories, the acquisition of visual fluency is socially mediated because visual fluency is a key constituent of the “visual language” students learn as part of discipline discourse. The acquisition of visual fluency involves learning to think, learn, and express oneself in terms of visual representations (Braden and Hortin 1981). Visual fluency allows students to infer community-specific ways of knowing and thinking beyond what is

explicitly depicted in the given visual representation (Airey and Linder 2009). The acquisition of visual fluency has been described as learning of a “visual vocabulary” that enables students to think about and communicate about discipline-specific phenomena in the same way that expert members of the community do (Schönborn and Anderson 2006).

Research on STEM suggests that visual fluency is an important aspect of students’ learning of content knowledge (Middendorf and Pace 2004; Northedge 2002, 2003; Tobias 1992) because it frees cognitive resources to engage in higher-order conceptual thinking (Gibson 2000; Goldstone and Barsalou 1998; Kellman and Garrigan 2009; Kellman and Massey 2013; Richman et al. 1996), to engage in creative thinking (NRC 2006; Richman et al. 1996), and to participate in discipline discourse (Braden and Hortin 1981; Northedge 2002, 2003; Taber 1998). Accordingly, visual fluency is considered an important educational goal in science (Kozma and Russell 2005; Wu and Shah 2004) and mathematics (Pape and Tchoshanov 2001).

In sum, cognitive learning theories and socio-cultural theories suggest learning *from* multiple visual representations involves visual fluency. Visual fluency is acquired via inductive learning processes that students engage in when using visual representations to participate in disciplinary discourse within scientific and professional communities.

Connectional Fluency: Translating Between Visual Representations

Cognitive learning theories suggest that fluency in making connections among visual representations is important for students’ learning from visual representations. Experts have the ability to “just see” whether two visual representations show the same information, to combine information from different visual representations without any perceived mental effort, and to quickly translate among them (Chase and Simon 1973; Kellman and Garrigan 2009; Kellman and Massey 2013). For example, a fractions expert might see “at a glance” that in scenario 3, the pie chart in Fig. 1b and the number line in Fig. 1c show the same principle of splitting six units into halves. That is, the fractions expert is highly efficient in making connections among visual representations. According to the CTML and ITCP (Mayer 2005, 2009; Schnotz 2005, 2014), these connections involve mapping the analog internal representations of the visual representations to one another. According to cognitive learning theories (Goldstone et al. 1997; Kellman and Massey 2013), students gain *efficiency* in making such mappings results from perceptual chunking: rather than mapping particular analog features to one another, students learn to treat the entire analog internal representation as one perceptual chunk. Hence, the acquisition of connectional fluency involves learning simple one-on-one mappings between perceptual chunks.

According to socio-cultural theories, the acquisition of connectional fluency is important because it enables students to participate in social interactions within scientific and professional communities that involve translating among multiple visual representations. For example, Airey and Linder (2009) describe how it becomes “almost second nature” for physics students to translate and flexibly switch from one visual representation to another (p. 10) and how making such connections allows students to infer community-specific knowledge across representations that goes beyond what is explicitly shown in the visual representations. Kozma et al. (2000) describe how expert chemists’ fluency in translating among multiple visual representations enables them to communicate with one another while solving complex problems collaboratively.

Like visual fluency, connectional fluency facilitates learning of content knowledge because it frees cognitive resources to engage in higher-order conceptual thinking about domain-

relevant concepts (Gibson 2000; Kellman and Massey 2013; Richman et al. 1996) and to participate in practices of scientific and professional communities (Airey and Linder 2009; Kozma and Russell 2005; Schönborn and Bögeholz 2013). Consequently, connectional fluency is considered to be an important learning goal in STEM disciplines (Deliyianni et al. 2015; Gilbert 2005, 2008; Gilbert and Treagust 2009; Hill 2015; NRC 2006).

In sum, cognitive learning theories and socio-cultural theories suggest learning *from* multiple visual representations involves connectional fluency. Connectional fluency is acquired via inductive learning processes that students engage in when translating among multiple visual representations in disciplinary discourse practices of scientific and professional communities.

Meta-Representational Competencies

A student asked to reflect on what he finds challenging about visual representations reported: “some [representations] can be difficult to find or there are multiple viable [representations] and I don’t know which is more favored.” This student seems to have difficulties in choosing appropriate visual representations for particular tasks. In fact, in many learning activities in STEM students have to choose which visual representation to use for a given task (NRC 2006).

Cognitive learning theories suggest that the set of competencies that allow students to make such choices are *meta-representational competencies*: knowledge and skills that enable students “to select, produce, and productively use representations, but also [...] to critique and modify representations” (diSessa and Sherin 2000), p. 386). Meta-representational competencies involve at least two rational considerations (Acevedo Nistal et al. 2015). First, the student needs to consider the task demands to find a representation that allows him/her to effectively and efficiently solve the task (Ainsworth 2006; Larkin and Simon 1987; Schnotz and Bannert 2003). Second, the student needs to consider his/her own ability to use the representation in the context of the task (Acevedo Nistal et al. 2013; Ainsworth 2006; Nistal et al. 2010a). Considering his/her own ability allows the student to choose a representation that matches demands of the given *task context* (e.g., to solve task 3 in Fig. 1c, a student may find that drawing a numberline on grid paper makes it easier to partition it into equally sized sections than drawing a pie chart), ability level (e.g., for task 3 in Fig. 1c, a numberline may be the most appropriate choice to depict distance, but the student is more familiar with pie charts and hence more likely to solve the task correctly with a pie chart), and goals (e.g., the student may have the goal to learn how to use numberline and will therefore choose the numberline over the pie chart).

Meta-representational competencies are a subcategory of meta-cognitive competencies. Meta-cognition involves knowing about one’s conceptual knowledge about the task and about the appropriateness of certain problem-solving strategies (Greene et al. 2012; Jacobs and Paris 1987), as well as expectations that a given strategy will lead to success (Bandura 2001). These considerations imply that meta-representational competencies require visual understanding: to choose an appropriate representation, students need to understand how the representation depicts domain-relevant information (diSessa and Sherin 2000; Nistal et al. 2011). Meta-representational competencies also build on connectional understanding (de Jong et al. 1998; Even 1990; Gagatsis and Elia 2004): to choose a suitable representation, students need to understand differences between multiple representations in what information they each depict (Ainsworth 2006; Cox 1999; Rau et al. 2015b).

Further, meta-cognition involves conditional knowledge about when a strategy is relevant to a task (Jacobs and Paris 1987; Winne and Azevedo 2014). This aspect of meta-cognition

allows students to take into account the context of problem solving when selecting an appropriate strategy (Hartman 2002; Pintrich 2003). These considerations also imply that meta-representational competencies require visual understanding: to choose the appropriate representation, students need to know which visual representation depicts information relevant to the task (Ainsworth 2006; Nistal et al. 2010b, 2011). Further, they involve knowledge about one's own ability level (Winne and Azevedo 2014), in particular, about one's own visual understanding and visual fluency (Uesaka and Manalo 2006).

Cognitive learning theories suggest that meta-representational competencies are mostly acquired via verbally mediated sense-making processes (Cox 1999; diSessa and Sherin 2000) that require explicit instruction (Cox and Brna 1995; diSessa and Sherin 2000; McKendree et al. 2002). In support of this notion, research shows that even though experts often make meta-representational decisions automatically, they can verbalize why they chose a particular visual representation (Kohl and Finkelstein 2008; Kozma and Russell 1997). Thus, a high degree of fluency in choosing visual representations may be indicative of highly practiced sense-making processes, rather than of inductive learning processes.

According to socio-cultural theories, meta-representational competencies involve knowing about historical and cultural conventions for choosing particular visual representations for a given task (Kozma et al. 2000; Rocke 2010). While some of these conventions may be learned via explicit sense-making processes, socio-cultural theories also discuss how students may inductively learn these conventions through exposure to community practices over the course of the enculturation process (Airey and Linder 2009; Schönborn and Anderson 2006).

In sum, cognitive learning theories and socio-cultural theories suggest that meta-representational competencies are involved in learning from visual representations. Meta-representational competencies are acquired via sense-making processes and inductive processes that students engage in by participating in practices of choosing visual representations for authentic tasks within scientific and professional communities.

Comparison of Single Versus Multiple Visual Representations

The previous section described several representational competencies that are considered key to students' ability to learn content knowledge from visual representations. It seems plausible that the complexity of these representational competencies and their role for students' learning of content knowledge depends on how many different types of visual representations a learning activity involves. This section presents an analysis of how these representational competencies are affected by the number of visual representations. To this end, the scenario in which students learn with T+SV representations and the scenario in which they learn with T+MV representations are contrasted. This analysis yields seven hypotheses for conditions under which T+MV representations may be more effective than T+SV representations, summarized in Table 2. This analysis is organized by the representational competencies just reviewed.

Conceptual Competencies for Single and Multiple Visual Representations

Hypothesis 1: Visual Understanding in T+SV Versus T+MV Scenarios

Recall that visual understanding describes the knowledge and skills involved in connecting visual representations to domain-relevant concepts. Students need to acquire visual

Table 2 Overview of seven hypotheses describing conditions under which text plus multiple types of visual representations (T+MV) may be more effective than text plus a single type of visual representation (T+SV), as well as evidence for each hypothesis

Hypothesis	Rationale	Evidence
1. Visual understanding T+MV scenarios can only be more effective than T+SV scenarios if students (1) have prior visual understanding of the given visual representations or (2) receive instructional support for visual understanding.	Visual understanding is a facilitator of students' learning in T+SV scenarios, but a <i>necessary prerequisite</i> for students' learning in T+MV scenarios.	–<i>Indirect</i>: The ability to form an accurate internal representation of a visual representation is a prerequisite for student's learning from the visual representation. –<i>Indirect</i>: Instructional support for visual understanding enhances students' benefit from T+SV representations (compared to text-only). –<i>Direct</i>: Instructional support for visual understanding enhances students' benefit from T+MV representations (compared to T+SV scenarios). –<i>Indirect</i>: Visual understanding enables students to make connections among multiple visual representations.
2. Connectional understanding An advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior connectional understanding for the given visual representations or (2) receive instructional support for connectional understanding.	Connectional understanding is not involved in learning in T+SV scenarios, but is a <i>facilitator</i> of students' learning in T+MV scenarios.	–<i>Indirect</i>: Failure to make connections among visual representation can interfere with students' learning of content knowledge. –<i>Indirect</i>: Connectional-understanding support enhances students' learning of content knowledge from T+MV representations. –<i>Direct</i>: T+MV representations are more effective than T+SV representations only when combined with connectional-understanding support. –<i>Indirect</i>: Connectional understanding is a prerequisite for students' acquisition of connectional fluency.
3. Visual fluency An advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior visual fluency, or (2) receive instructional support for visual fluency.	Visual fluency may play a more important role as a <i>facilitator</i> of students' learning in T+MV than in T+SV scenarios.	–<i>Indirect</i>: Students who lack visual fluency tend to focus on surface features and engage in ineffective visual search processes while learning with T+SV and T+MV representations. –<i>Indirect</i>: Students with visual fluency more readily make connections to novel visual representations. –<i>Indirect</i>: Visual-fluency support enhances students' learning of content knowledge from visual representations.
4. Connectional fluency An advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior connectional fluency for the given visual representations or (2) receive instructional	Connectional fluency is not involved in learning in T+SV scenarios, but is a <i>facilitator</i> of students' learning in T+MV scenarios.	–<i>Indirect</i>: Connectional-fluency support enhances students' learning from T+MV representations (compared T+MV representations without connectional-fluency support). –<i>Direct</i>: MV representations are more effective than T+SV representations only when combined with connectional-fluency support.

Table 2 (continued)

Hypothesis	Rationale	Evidence
support for connectational fluency.		– <i>Indirect</i> : Connectational-fluency support complements connectational-understanding support.
5. Meta-representational competencies An advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior meta-representational competencies, or (2) receive instructional support for meta-representational competencies.	Because meta-representational decisions are more complex in T+MV than in T+SV scenarios, the role of meta-representational competencies is more important in T+MV than in T+SV scenarios.	– <i>Indirect</i> : Lack of meta-representational competencies can result in inappropriate choices of visual representations, which interferes with students' learning of content knowledge. – <i>Indirect</i> : Instructional support for meta-representational competencies enhances students' learning outcomes in T+MV scenarios.
6. Information overlap T+MV scenarios can only be more effective than T+SV scenarios if each visual representation provides information about the learning content that (1) adds relevant information not fully covered by the other visual representations but (2) overlaps partially with information provided in the other visual representations.	Information from irrelevant representations or unrelated visual representations cannot be integrated into a correct mental model; integrating information from redundant visual representations is cognitively demanding but does not yield additional insight.	– <i>Indirect</i> : T+SV representations are only more effective than text-only if the visual representation provides relevant information. – <i>Indirect</i> : Many studies that fail to show an advantage of T+MV over T+SV representations included multiple visual representations that did not differ on design dimensions (e.g., abstract, virtual). – <i>Indirect</i> : T+MV representations that differ on design dimensions (e.g., concrete/abstract, physical/virtual) are more effective than T+MV representations that do not differ on these dimensions.
7. Socio-cultural conventions T+MV scenarios are more effective at enculturating students into community practices than T+SV scenarios if the use of each visual representation aligns with its use in the target community of practice.	The goal of education is to enable students to participate in practices common in scientific and professional communities, which involves the use of visual representations.	– <i>Indirect</i> : The role of representational competencies in education differs between domains, and these differences reflect differences in socio-cultural practices of the use of visual representations in these domains.

understanding for *each* visual representation. Therefore, one difference is that students need visual understanding for a larger number of representations in T+MV than in T+SV scenarios.

In addition, visual understanding plays a qualitatively different role for students' learning success in T+SV versus T+MV scenarios. In *T+SV scenarios*, visual understanding helps students connect the visual representation to text because students do not have to search the visual representation for relevant features. Yet, connecting visual representations to text is also a major mechanism through which students acquire visual understanding. Hence, visual understanding is not a prerequisite but a facilitator of students' learning from T+SV representations.

In *T+MV scenarios*, by contrast, students have to make not only text-visual connections, but they also have to learn to make visual-visual connections (i.e., connectional understanding and connectional fluency). Above, it was argued that different visual representations are more likely to have similar mappings to the referent than text and visual representations, and therefore, surface similarities may be more misleading when students learn to make visual-visual connections than visual-text connections. This factor may be corroborated by the fact that the presence of visual representations tends to reduce students' attention to text (Mayer and Gallini 1990; Schnotz 2014). If text serves less of a guiding role, visual features serve a larger role in guiding students' connection making in *T+MV* than in *T+SV* scenarios. Unless students know which visual features are conceptually relevant (i.e., visual understanding), they may rely on surface features to make connections, which may result in failure to make correct connections. Therefore, because visual understanding equips students with the knowledge necessary to distinguish relevant features from surface features, visual understanding may be a *necessary prerequisite* for students' acquisition of connectional understanding, and, hence, an important prerequisite for students' learning in *T+MV* scenarios. This analysis yields *hypothesis 1*: *T+MV* scenarios can only be more effective than *T+SV* scenarios if students (1) have prior visual understanding of the given visual representations (or of a different visual representation with equivalent feature-referent mappings that transfers to the given visual representation; see Gentner et al. 2003) or (2) receive instructional support for visual understanding.

Hypothesis 2: Connectional Understanding in T+MV Scenarios

Students have to make connections among multiple types of visual representations in *T+MV* scenarios, but not in *T+SV* scenarios. According to Ainsworth (2006, 2014), students can acquire connectional understanding while they learn content knowledge—in fact, connection-making processes are considered to be a major mechanism through which students refine their mental model of the content (Ainsworth 2006, 2014; Seufert 2003). However, connection making requires cognitive resources. Consequently, students with connectional understanding should experience lower cognitive load while learning content knowledge. Therefore, connectional understanding may be a *facilitator* of students' learning of content knowledge. This analysis suggests *hypothesis 2*: an advantage of *T+MV* scenarios over *T+SV* scenarios is more pronounced if students (1) have prior connectional understanding for the visual representations (or transfer connectional understanding from visual representations with similar feature-referent mappings; Gentner et al. 2003), or (2) receive instructional support for connectional understanding.

Perceptual Competencies for Single and Multiple Visual Representations

Hypothesis 3: Visual Fluency in T+SV Versus T+MV Scenarios

The literature reviewed above suggests that visual fluency is an important aspect of students' learning of content knowledge. Students need to acquire visual fluency with each of the visual representations. Because *T+MV* scenarios include more visual representations than *T+SV* scenarios, students need to acquire visual fluency for a larger number of visual representations.

Furthermore, the role of visual fluency for students' learning success may be different for *T+SV* and *T+MV* scenarios because students have to make visual-visual connections only in

T+MV scenarios. Above, it was argued that connection making among visual representations requires that students know which visual features are conceptually relevant (visual understanding). A similar argument can be made for visual fluency. Because visual fluency allows students to quickly and effortlessly see the concepts shown in visual representations, it frees cognitive resources for higher-order thinking. Because the acquisition of connectional understanding is cognitively demanding and difficult, fluency in processing the visual representation should facilitate students' acquisition of connectional understanding. This factor is corroborated by the fact that text may serve less of a guiding role in T+MV scenarios. Therefore, visual fluency may play a more important role as a *facilitator* of students' learning in T+MV than in T+SV scenarios. This yields *hypothesis 3*: an advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior visual fluency, or (2) receive instructional support for visual fluency.

Hypothesis 4: Connectional Fluency in T+MV Scenarios

In T+MV scenarios (but not in T+SV scenarios), students have to acquire connectional fluency. Recall that connectional fluency is acquired via inductive learning processes that occur automatically when students are repeatedly exposed to visual representations. Because inductive learning processes are implicit and do not require effortful, explicit engagement, the acquisition of connectional fluency does compete with cognitive resources that students need to learn content knowledge. Hence, connectional fluency is not a prerequisite for learning of content knowledge. However, because it frees cognitive resources for higher-level thinking, it *facilitates* learning of content knowledge. This suggests *hypothesis 4*: an advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior connectional fluency for the given visual representations or (2) receive instructional support for connectional fluency.

Meta-Representational Competencies

Hypothesis 5: Meta-Representational Competencies in T+SV Versus T+MV Scenarios

Meta-representational competencies describe the knowledge and skills that enable students to critically reflect on advantages and limitations of a representation for a given task and to use this reflection to choose an appropriate representation to solve the task. The T+SV and T+MV scenarios differ with respect to the complexity of meta-representational decisions a student has to make. In *T+SV scenarios*, there is only one type of visual representation. Hence, the student has to decide whether and to what extent to use it. For example, in Fig. 1b, the student may choose to not attend to the pie chart at all, to view it as an illustration of the task, or to manipulate it to solve the task. To make this decision, the student needs to consider whether the visual representation matches the task, which involves visual understanding. Further, the student needs to consider whether using the visual representation increases his/her chances of correctly solving the task. This decision involves visual understanding, and the student's knowledge about his/her own visual understanding, and visual fluency. Thus, in the T+SV representation scenario, meta-representational competencies involve both visual understanding and visual fluency.

In *T+MV scenarios*, the student has to decide whether to use any visual representation to solve the given task, and—if so—which visual representation to choose (e.g., in Fig. 1c, the pie chart, rectangle, or numberline). This decision involves connectional understanding: the

student needs to consider similarities (e.g., the pie chart and the numberline both show fractions as proportions of some quantity that is partitioned in one dimension) and differences between the visual representations (e.g., the numberline makes it easier to show fractions larger than 1). While connectional understanding with respect to similarities allows the student to establish that a set of visual representations are appropriate choices, connectional understanding with respect to differences allows him/her to identify pros and cons of using one visual representation over the other. It is conceivable that connectional fluency (i.e., the ease with which the student can make these connections) facilitates this decision-making process. Hence, for T+MV scenarios, meta-representational competencies involve connectional understanding, connectional fluency, visual understanding, and visual fluency. This means that meta-representational competencies are more complex in T+MV than in T+SV scenarios. If meta-representational competencies are crucial to students' learning from visual representations, the fact that they are more complex implies that they are more important for students' learning from T+MV than from T+SV representations. This analysis suggests *hypothesis 5*: An advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior meta-representational competencies, or (2) receive instructional support for meta-representational competencies.

Additional Hypotheses Deduced from Cost-Benefit Analysis

Thus far, this analysis suggests that students' benefit from T+MV scenarios may depend on their acquisition of particular representational competencies. The cognitive learning theories and socio-cultural theories reviewed above suggest that students acquire these competencies via sense-making processes and inductive learning processes that are socially mediated. Sense-making processes require significant cognitive effort because students need to willfully engage in them (Fiorella and Mayer 2015; Koedinger et al. 2012; Wylie and Chi 2014), which often requires additional instructional time for explicit support (Ainsworth et al. 2002; Bodemer and Faust 2006; Rau et al. 2014). Inductive learning processes are not necessarily cognitively demanding because they are implicit. However, they do require significant instructional time because they rely on students' experiences with numerous examples and extensive practice (Kellman and Massey 2013). Thus, the acquisition of these representational competencies comes at the cost of significant cognitive effort and instructional time. This consideration is particularly important in cases where we cannot assume that students may transfer representational competencies from similar visual representations (Gentner et al. 2003) because visual representations have different feature-referent mappings or because visual and connectional fluency tend to be specific to a type of visual representation (Ackerman 2003; Gibson 1969, 2000).

Consequently, T+MV scenarios can only be more effective than T+SV scenarios if each of the visual representations is "worth the costs" (Ainsworth 2006, 2014; Schnotz 2014). Multimedia learning theories and socio-cultural theories yield additional hypotheses as to when the benefits of an additional visual representation outweigh its costs.

Hypothesis 6: Multimedia Learning Theories on Information Overlap

Multimedia learning theory suggests that students' learning of content knowledge only benefits from a visual representation if it adds novel information that is relevant to the to-be-learned content. If the visual representation does not add useful information, the attempt to

integrate the visual representation with prior knowledge and/or with text is unproductive in the sense that it is cognitively demanding but does not add to the student's understanding of the content (Carney and Levin 2002; Schnotz and Bannert 2003). This effect applies to T+SV as well as to T+MV situations, but it may be compounded for T+MV situations because *each* additional visual representation needs to add relevant information to the learning task that is not fully captured by the other visual representations (Ainsworth 2006, 2014). A further consideration for T+MV scenarios is the *information overlap* among multiple visual representations (Carney and Levin 2002). On the one extreme, the information overlap between two visual representations may be zero; that means there are no connections to be made. In this case, the student may attempt to make connections among the visual representations. The student may fail to make connections (because there are none to be made), which may unnecessarily increase learning time. Alternatively, the student may make incorrect connections, which may decrease learning outcomes. On the other extreme, the information overlap between two visual representations may be extremely high; that means the two visual representations present virtually the same information (i.e., they are redundant). In this case, the student may not benefit from making connections because one visual representation presents all the necessary information, and the cognitive demands that result from connection making simply do not “buy” the student any additional insight.

This analysis suggests *hypothesis 6*: T+MV scenarios can only be more effective than T+SV scenarios if each visual representation provides information about the learning content that (1) adds relevant information not fully covered by the other visual representations but (2) overlaps partially with information provided in the other visual representations.

Hypothesis 7: Socio-Cultural Theories on Conventions for Visual Representations

From a socio-cultural perspective on learning, the goal of education is to enculturate students into community practices, that is, to enable students to participate in practices common in scientific and professional communities. These practices evolved as implicit conventions that are historically rooted and as explicit consensus about the usefulness of representations for real-world tasks or for instruction. Therefore, the use of visual representations in learning activities should align with the implicit and explicit conventions for using visual representations. For example, if expert members in the target community use certain visual representations, learning activities should expose students to the same visual representations. In some cases, that implies there may be value in presenting students with a visual representation purely because experts use the visual representation and students need to learn to use it if they are to become legitimate members in the target communities of practice, or because of the representation's historical impact on the community. This analysis suggests *hypothesis 7*: T+MV scenarios are more effective at enculturating students into community practices than T+SV scenarios if the use of each visual representation aligns with its use in the target community of practice.

Evidence for Conditions for Effectiveness of Multiple Visual Representations

This section examines empirical evidence for each of these seven hypotheses, which allows addressing the main research question of this paper: under which conditions T+MV scenarios might be more effective than T+SV scenarios for students' learning of content knowledge?

Evidence for these hypotheses is considered to be direct if studies experimentally tested the hypothesis, which requires a comparison of T+SV and T+MV scenarios. Evidence is considered to be indirect if studies yield results that are consistent with the hypothesis without experimentally testing it. Table 2 lists the evidence for each hypothesis.

Hypothesis 1: Visual Understanding

Based on the rationale that visual understanding is a facilitator of learning in T+SV scenarios, but a prerequisite for learning in T+MV scenarios, hypothesis 1 suggests that T+MV scenarios are more effective than T+SV scenarios if students (1) have prior visual understanding of the given visual representations or (2) receive instructional support for visual understanding.

Indirect evidence for this hypothesis comes from studies that have manipulated students' abilities to form accurate internal representations of visual representations. An experiment that compared learning text-only to learning with T+SV representations, Eitel et al. (2013) varied when the visual representation was presented: concurrently, or at various system-controlled time-lags, or student-controlled time-lags before the text (i.e., the visual representation would disappear when the text showed up). Results showed that students in the T+SV condition outperformed the text-only condition if the visual representation was presented concurrently or with a student-controlled time-lag, but not with a system-controlled time-lag. It seems plausible that shorter time-lags interfered with students' ability to form an accurate internal representation. Because visual understanding involves the ability to form an accurate internal representation of a visual representation, this study suggests that visual understanding is indeed a prerequisite for student's learning of content knowledge from the visual representation.

Additional indirect evidence for hypothesis 1 comes from studies on instructional support for visual understanding that helps students map visual representation to domain-relevant concepts, for example via self-explanation prompts. Research on support for visual understanding builds on observations that students tend not to spontaneously make connections between visual representations and text (Ainsworth et al. 2002; Bodemer and Faust 2006; Rau et al. 2014). For example, in Fig. 1b, students may fail to notice that the pie chart relates to the text-based sharing scenarios. Studies on T+SV scenarios show that prompting students to make connections between text and the visual representation enhances students' learning outcomes (Bodemer and Faust 2006; Seufert 2003; Van der Meij and de Jong 2011). However, even when prompted, students may not know how to make correct connections. Research on T+SV scenarios shows that students' learning can be enhanced by providing instructional support that helps them connect conceptually relevant features of the visual representation to corresponding parts of text. For example, (Baetge and Seufert 2010) investigated the effects of instructional support for visual understanding that helped students connect the visual representation to text via color coding and hyperlinking. They compared text-only, T+SV without visual-understanding support, and T+SV with visual-understanding support. They found that T+SV led to higher learning outcomes than text-only provided that students received visual-understanding support. Berthold and Renkl (2009) also investigated the effects of visual-understanding support, provided in the form of color coding that helped students connect the visual representation to text. They compared text-only, T+SV without visual-understanding support, and T+SV with support. The results showed that there was an advantage of learning with T+SV only when combined with visual-understanding support. Therefore, students' benefit from T+SV seems to depend on their ability to make correct connections between the visual representation and text.

Direct evidence for this hypothesis comes from experiments that compared T+SV and T+MV scenarios. Rau and colleagues (2015a) compared the effects of T+SV (numberlines only) and T+MV representations (pie charts, rectangles, numberlines) on students' learning of fractions knowledge. A second experimental factor was whether or not students received visual-understanding support, provided in the form of self-explanation prompts to connect each visual representation to text-based representations. Results showed that T+MV representations were more effective than T+SV representations *only if* students received visual-understanding support. For students who did not receive visual-understanding support, T+SV representations were more effective than T+MV representations. This finding aligns with the notion that visual understanding is not only a facilitator, but a prerequisite for students' learning in T+MV scenarios.

Finally, indirect evidence for hypothesis 1 comes from research that examined the role of visual understanding for connection making among visual representations in T+MV scenarios. For example, in a study that combined eye-tracking and interviews, Cook et al. (2007) found that students with low prior content knowledge tend to focus on surface features when they make connections, whereas students with high prior content knowledge tend to focus on conceptually relevant features. It is plausible that students with high prior content knowledge had higher visual understanding. Furthermore, a think-aloud study by Rau and colleagues (2014) showed that students tend to make surface-level connections among visual representations unless they received instructional support for visual understanding, provided in the form of directed prompts to focus on conceptually relevant features.

In summary, while most of the evidence is indirect, there is emerging direct evidence in support of hypothesis 1. Taken together, the reviewed studies suggest that T+MV scenarios can only be more effective than T+SV scenarios if students have previously acquired or receive instructional support for visual understanding for each visual representation. Furthermore, these studies illustrate that visual-understanding support can help students acquire visual understanding while they learn content knowledge. Therefore, we may, for example, expect that the T+MV scenario in Fig. 1c is more effective than the T+SV scenario in Fig. 1b if students receive support to reason about how the pie chart, rectangle, and numberline representations show key concepts such as numerator and denominator—for instance via self-explanation prompts.

Hypothesis 2: Connectional Understanding

Based on the argument that connectional understanding facilitates learning in T+MV scenarios (but not in T+SV scenarios), hypothesis 2 suggests that an advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior connectional understanding or (2) receive instructional support for connectional understanding.

Indirect evidence for this hypothesis comes from research showing that failure to make connections among visual representation can interfere with students' learning of content knowledge from T+MV representations (Ainsworth et al. 2002; Bodemer and Faust 2006; Rau et al. 2014). Further indirect evidence for this hypothesis comes from research that has investigated the effects of instructional support for connectional understanding in T+MV scenarios. Such connectional-understanding support typically asks students to explain that visual features of different visual representations show corresponding or complementary information about the same concept. A number of experiments show that providing connectional-understanding support enhances students' learning of content knowledge. For example, Van der Meij and de Jong (2011) tested the effects of connectional-understanding support that prompted students to

self-explain connections among multiple visual representations. They compared students who learned with T+MV representations with such connectional-understanding support to students who learned with T+MV representations without such support. Results showed higher learning outcomes on a content knowledge test for students who received connectional-understanding support. Gutwill et al. (1999) tested the effects of connectional-understanding support that provided step-by-step guidance to find corresponding features in different visual representations. Students with connectional-understanding support showed higher learning of content knowledge than students in a control group without connectional-understanding support. In a qualitative study, Linenberger and Bretz (2012) provided connectional-understanding support via teacher questions that asked students to explain differences between visual representations. Based on an analysis of interviews and student-teacher dialogue, they argue that connectional-understanding support enhanced students' learning of the content knowledge.

Further, direct evidence for hypothesis 2 comes from experiments showing that students learning with T+MV representations outperform students learning with T+SV representations only if they receive connectional-understanding support. Rau et al. (2012) compared students learning with T+MV representations who received connectional-understanding support in the form of self-explanation prompts to students learning with T+MV representations without such support and to students learning with T+SV representations. Results showed that only students who received connectional-understanding support outperformed the T+SV condition.

Finally, indirect evidence for hypothesis 2 comes from research showing that connectional understanding is a prerequisite for students' acquisition of other representational competencies that facilitate students' learning of content knowledge, specifically of connectional fluency. Rau and colleagues (2013b; Rau, Making connections in chemistry: How does learning to explicitly compare visual representations relate to learning to automatically classify representations?, under review) tested different sequences of instructional support for connectional understanding and for connectional fluency. Students who received connectional-understanding support benefited more from subsequent connectional-fluency support than students who had not previously received connectional-understanding support. Further, students with low prior connectional understanding did not benefit from connectional-fluency support unless they also received connectional-understanding support (Rau, Making connections in chemistry: How does learning to explicitly compare visual representations relate to learning to automatically classify representations?, under review).

In summary, there is mostly indirect and some direct evidence for hypothesis 2. Taken together, research suggests that T+MV scenarios are more effective than T+SV scenarios particularly if students have acquired or receive instructional support for connectional understanding. Further, students can acquire connectional understanding while they learn content knowledge if they receive instructional support to make sense of the connections. For example, we may expect that the T+MV scenario (Fig. 1c) is more effective than the T+SV scenario (Fig. 1b) if students receive support to reason about which visual features of the pie chart correspond to which visual features in the rectangle and the numberline.

Hypothesis 3: Visual Fluency

Based on the rationale that visual fluency may play a more important role as a facilitator of students' learning in T+MV than in T+SV scenarios, hypothesis 3 suggests that an advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior visual fluency, or (2) receive instructional support for visual fluency.

Indirect empirical evidence for hypothesis 3 comes from studies on students who lack visual fluency in T+SV and T+MV scenarios. Even if students have visual understanding, working with them may require considerable cognitive effort and may involve ineffective visual search processes. Eye-tracking studies indicate that visually fluent students are more efficient at allocating visual attention to relevant visual features (for an overview, see Gegenfurtner et al. 2011). For example, Carmichael et al. (2010) found that experts direct their attention to conceptually relevant features of visual representations, whereas novices tend to focus on salient surface features. These eye-gaze measures correlated with students' performance in using the visual representations to solve domain-relevant tasks.

Further indirect support for hypothesis 3 comes from research on connection making among visual representations. Recall that visual fluency frees cognitive resources that students can invest into connection making among visual representations. Hinze et al. (2013a) investigated whether fluency with a familiar visual representation affects adoption of a new visual representation. Eye-gaze data showed that students who are fluent with the familiar visual representation more readily adopted the new visual representation. Arguably, students learned about the new visual representation by connecting it to the familiar visual representation.

Finally, indirect support for hypothesis 3 comes from studies on instructional support for visual fluency in T+MV scenarios. Such instructional support exposes students to numerous example representations designed so that students are likely to induce which visual features are relevant, for example by providing contrasting cases (Massey et al. 2011) or by interleaving relevant features (Rau et al. 2014). Kellman and colleagues developed visual-fluency supports that provide students with many short tasks that ask them to rapidly translate between multiple external representations. For example, a student might be asked to select one of several pie charts that correspond to a word problem. The task sequences are designed to expose students to systematic variation, often in the form of contrasting cases, so that irrelevant features vary but relevant features appear across several tasks (Kellman and Massey 2013; Massey et al. 2011). Kellman and colleagues evaluated this type of visual-fluency support in a number of controlled experiments and observational studies in T+MV scenarios for several domains, including algebra, fractions, and chemistry (Kellman et al. 2009; Kellman et al. 2008; Wise et al. 2000). Results showed that visual-fluency support led to large and lasting gains on tests of visual fluency. Furthermore, gains in visual fluency were associated with higher learning outcomes on content knowledge tests. Other forms of visual-fluency support, such as interleaving sequences of visual representations so that students encounter a different type of visual representation in consecutive tasks (Rau et al. 2014) and use them for different task types (Rau et al. 2013a), have also been shown to enhance learning of content knowledge.

In sum, there is indirect evidence in support of hypothesis 3, which suggests that T+MV scenarios are more effective than T+SV scenarios particularly if students have acquired or receive instructional support for visual fluency. While students can acquire visual fluency through extensive practice, targeted trainings have also shown to be effective. Therefore, we may for example expect that a student who is fluent with pie charts but not with rectangles and numberlines might benefit more from the T+SV scenario (Fig. 1b) than from the T+MV scenario (Fig. 1c) if he/she receives additional support for visual fluency with rectangles and numberlines. However, a direct experimental test of hypothesis 3 is still outstanding.

Hypothesis 4: Connectional Fluency

Based on the argument that connectional fluency is a facilitator of students' learning in T+MV scenarios (but not in T+SV scenarios), hypothesis 4 suggests that an advantage of T+MV

scenarios over T+SV scenarios is more pronounced if students (1) have prior connectional fluency for the visual representations or (2) receive instructional support for connectional fluency.

Indirect evidence for hypothesis 4 comes from studies by Kellman and colleagues who developed connectional-fluency supports that ask students to rapidly translate among visual representations (Kellman et al. 2009; Massey et al. 2011; Wise et al. 2000). For example, students may select one of several numberlines that show the same fraction as a pie chart. Experiments show that such connectional-fluency supports enhance learning of content knowledge in a variety of domains (Kellman et al. 2009; Kellman et al. 2008; Wise et al. 2000).

Direct evidence for hypothesis 4 comes from an experiment that compared learning with T+MV representations with or without connectional-fluency support to learning with T+SV representations (Rau et al. 2012). In this study, students learned about fractions using pie charts, rectangles, and numberlines. Only students who received connectional-fluency support showed higher learning gains in content knowledge than the T+SV condition. Follow-up studies suggested that connectional understanding and connectional fluency are indeed separate representational competencies that are best supported by different types of instructional interventions. Rau et al. (2012) and Rau and Wu (2015) found that students who received both connectional-fluency support (designed to engage nonverbal inductive learning processes) and connectional-understanding support (designed to engage verbally mediated sense-making processes) showed higher learning gains on content knowledge tests than students who received either type of support alone.

In sum, although most of the evidence for hypothesis 4 is indirect, there is some emerging direct evidence in support of hypothesis 4. Taken together, this evidence suggests that T+MV scenarios are more effective than T+SV scenarios particularly if students have acquired or receive instructional support for connectional fluency. While students can acquire connectional fluency through extensive practice in translating across multiple visual representations, targeted trainings have also shown to be effective. Therefore, we may, for example expect that the T+MV scenario (Fig. 1c) is more effective than the T+SV scenario (Fig. 1b) if students are fluent in translating among the pie chart, rectangle, and numberline representations.

Hypothesis 5: Meta-Representational Competencies

Based on the argument that the higher complexity of meta-representational decisions in T+MV than in T+SV scenarios implies that the role of meta-representational competencies is more important in T+MV than in T+SV scenarios, hypothesis 5 states that an advantage of T+MV scenarios over T+SV scenarios is more pronounced if students (1) have prior meta-representational competencies, or (2) receive instructional support for them.

Indirect evidence for this hypothesis comes from studies showing that choosing inappropriate visual representations for a learning activity may jeopardize the students' learning of content knowledge (Gruber et al. 1995; Schnotz and Bannert 2003; Yerushamly 1991). Therefore, if students choose their visual representations themselves, making inappropriate choices may equally jeopardize their learning success. Students' difficulties in choosing visual representations are well documented: students vary largely in their ability to select (Roberts et al. 1997) or construct (Cox and Brna 1995) visual representations that are appropriate for a task, and they often fail to select the appropriate visual representation to solve a given task (Uesaka and Manalo 2006).

Additional indirect evidence comes from research on instructional support for meta-representational competencies in T+MV scenarios. For example, Uesaka and Manalo (2006) showed that asking students to compare visual representations resulted in more appropriate representational choices and higher learning outcomes on a content knowledge test, compared to a control condition without such support. Grawemeyer (2006) evaluated an educational technology that supports students in choosing appropriate visual representations for a given task. Results showed that students improved their performance on a content knowledge test. Acevedo Nistal et al. (2015) tested a type of support that helped students match representational choices to task demands and to personal characteristics. Students who received this type of support made more adaptive representational choices and showed higher outcomes on a content knowledge test, compared to a control group without such support. However, because these studies did not involve a T+SV control condition, they did not directly test hypothesis 5.

In sum, there is indirect evidence for hypothesis 5, suggesting that an advantage of T+MV over T+SV scenarios is more pronounced if students have acquired or receive instructional support for meta-representational competencies. For example, the T+MV scenario (Fig. 1c) may be more effective than the T+SV scenario (Fig. 1b) if a student knows to select the pie chart for task 1 because it matches the pizza scenario and the numberline for task 3 because it matches the distance scenario. However, a formal test of his hypothesis 5 is outstanding.

Hypothesis 6: Information Overlap

Above, it was argued that information from irrelevant or unrelated visual representations cannot be integrated into a correct mental model, and that integrating information from redundant visual representations is cognitively demanding but does not yield additional insight. Based on this argument, hypothesis 6 suggests that an advantage of T+MV scenarios over T+SV scenarios is more pronounced if each visual representation provides information about the learning content that (1) adds relevant information not fully covered by the other visual representations but (2) overlaps partially with information provided in the other visual representations.

Indirect evidence for this claim comes from studies on T+SV scenarios that have considered whether the visual representation provides information that is relevant to the learning task (e.g., Carney and Levin 2002; Schnotz and Bannert 2003). For example, Schnotz and Bannert (2003) compared a text-only condition to two T+SV representations conditions in which the visual representation either did or did not align with the demands of the learning task. T+SV representations resulted in higher learning outcomes than text-only if the visual representation aligned with the task demands. Otherwise, the visual representation interfered with learning of content knowledge. Further, a study showing that T+SV representations can have a negative effect for students if the visual representation shows irrelevant information (Magner et al. 2012) can be interpreted as support for hypothesis 6. In this study, negative effects of irrelevant visual representations were limited to low-prior knowledge students. It is possible that low-prior knowledge students did not realize that the visual representations were not useful, and hence tried to attempt to integrate them with the text. Even though these studies were conducted in T+SV scenarios, these effects should generalize to T+MV scenarios.

Further indirect evidence for hypothesis 6 comes from research on information overlap among multiple visual representations in T+MV scenarios. Studies by Ainsworth et al. (2002) and by Bodner and Domin (2000) found that T+MV representations did not result in higher learning outcomes than T+SV representations if one visual representation provided all necessary information. Considering the variability of visual representations in experiments that

failed to show an advantage of T+MV representations over T+SV representations can also be interpreted as supporting hypothesis 6 (Berthold and Renkl 2009; Corradi et al. 2012; Rau et al. 2012, 2015a). In these studies, visual representations in the T+MV conditions did not differ on dimensions of concreteness or interactivity; all visual representations were abstract and virtual. It is possible that visual representations that differ on these dimensions have less information overlap than visual representations that do not. Hence, multiple visual representations that differ on these design dimensions may have a higher pay-off than multiple visual representations that do not differ on these dimensions (Carney and Levin 2002). Indeed, there is considerable evidence for the effectiveness of combining visual representations that differ on these dimensions. For example, combining concrete and abstract visual representations (Braithwaite and Goldstone 2013; Fyfe et al. 2014; McNeil and Fyfe 2012; Moreno et al. 2011), and combining physical and virtual representations (Olympiou and Zacharia 2012; Zacharia et al. 2008) was shown to be more effective than T+MV representations that were only abstract, only concrete, only physical, or only virtual. Yet, because these studies did not include a T+SV control condition, they did not directly test hypothesis 6.

In sum, there is indirect evidence in support of hypothesis 6, suggesting that T+MV scenarios are more effective than T+SV scenarios if each added visual representation contributes information that is relevant to the learning task and overlaps to a medium degree with information shown in other visual representations. For example, we may expect the T+MV scenario (Fig. 1c) to be more effective than the T+SV scenario (Fig. 1b) only if the numberline provides information about the sharing tasks that is not captured by the pie chart and the rectangle. However, a direct test of hypothesis 6 is still outstanding.

Hypothesis 7: Socio-Cultural Conventions

Based on the socio-cultural perspective that the goal of education is to enable students to participate in practices common in scientific and professional communities, hypothesis 7 suggests that T+MV scenarios are more effective than T+SV scenarios if the use of each visual representation aligns with its use in the target community of practice. Specifically, this perspective suggests that we need to consider how social and cultural conventions affect representational competencies (Rau et al. 2015b). In particular, we may need to consider the following:

- Which visual representations students are exposed to (i.e., the fact that students use multiple visual representations in the first place)
- How the visual representations are to be interpreted (i.e., visual understanding)
- What level of fluency in using the representation is desirable and how it is to be assessed (i.e., visual fluency)
- What connections students are asked to make among specific visual representations (i.e., connectional understanding)
- To what extent it is necessary to be able to fluently translate among specific visual representations (i.e., connectional fluency)
- Which visual representations are deemed to be suitable for which types of tasks, and whether there are predicated preferences for one visual representation over the other (i.e., meta-representational competencies)

To the best of my knowledge, there is no empirical research that explicitly tested hypothesis 7. Yet, indirect evidence can be obtained by comparing the use of visual representations in

different domains. A comparison of two example domains illustrates this point: these domains fundamentally differ in how multiple visual representations are used in expert communities, which has implications for the representational competencies students need to acquire.

Fractions in Mathematics: Promoting Abstract Understanding

Fractions are an inherently difficult topic for elementary school students because they encompass many concepts, including parts of a whole (Cramer and Henry 2002; Cramer and Wyberg 2009), measurement (Carpenter 1971), ratios (Fuson and Abrahamson 2005; Lamon 1999), equi-partitioning (Confrey and Maloney 2010), as well as units and re-unitizing (Cramer and Henry 2002; Cramer and Wyberg 2009; Lamon 1999; Yanik et al. 2008). Different visual representations emphasize different conceptual interpretations of fractions (Charalambous and Pitta-Pantazi 2007). Commonly used visual representations of fractions include area models that help students understand equi-partitioning and parts-of-a-whole concepts (e.g., pie chart, rectangles), linear models that highlight measurement concepts (e.g., fraction strips, numberlines), and discrete models that emphasize ratio interpretations (e.g., sets, counters). Instruction often starts out by using concrete representations (e.g., pictures of pizzas) and physical representations (e.g., real pizzas) in order to ground student's visual understanding in concrete experiences (e.g., sharing activities; (Cramer and Wyberg 2009; Moss 2005; Proctor et al. 2002).

Connectional understanding is considered key to students' fractions learning (NCTM 2000, 2006; NMAP 2008; Siegler et al. 2010; Thompson and Saldanha 2003). Hence, instruction typically supports students in making connections by mapping conceptually relevant features across representations (e.g., by mapping the numerator shown by a shaded section in a pie chart to the number of sections between 0 and a dot on a numberline). Ultimately, the goal of connection making is to help students gain abstract understanding of fractions as proportions of a unit and to promote understanding of the symbolic notation of fractions and decimals (Hecht et al. 2007; Miura and Yamagishi 2002). Indeed, supports for promoting connectional fluency have been shown to enhance students' learning of fractions (Kellman et al. 2008; Rau et al. 2012). However, connectional fluency serves as a means to an end in enhancing students' implicit understanding of abstract proportions, rather than being an end in and of itself. In the same way, meta-representational competencies play less of a role in fractions expertise. Students may learn that a pie chart is more appropriate than a numberline when it comes to depicting sharing scenarios. Yet, this is not a decision a mathematics expert typically faces; mathematics experts typically rely on symbols (Kieren 1993) and one dominant visual representation: coordinate systems (Even 1998; Janvier et al. 1993), which build on knowledge about numberline representations (Siegler et al. 2010).

Thus, fractions is a domain in which the processes involved in learning with T+MV representations is mainly instrumental to enhancing students' learning of abstract concepts so that they can come to rely on symbols and coordinate systems when they become experts.

Bonding in Chemistry: Communicating About Concepts

Students struggle with crucial chemistry concepts related to bonding across high school, undergraduate, and graduate levels (Cooper et al. 2012; Nicoll 2001; Talanquer 2006). These difficulties result from the fact that many key concepts cannot be observed with the regular eye (Davidowitz and Chittleborough 2009) and are inherently abstract (Justi and Gilbert 2002). Different visual representations provide complementary views on these concepts (Bodner and Domin 2000; Coll and Treagust 2003; Kozma and Russell 2005). Relying on only one visual

representation can severely interfere with students' learning (Furio et al. 2000; Gabel and Bunce 1994). For example, to understand how atoms bond to form molecules, students typically encounter the visual representations shown in Fig. 2: they need to understand which atoms participate in the bonds (Lewis structures), bond angles and molecular volume (ball-and-stick models and space-filling models), and the distribution of electrons across the molecule (electronic potential maps) (Bowen 1990; Hinze et al. 2013b; Shusterman and Shusterman 1997). Chemistry experts have sophisticated mental models of which visual representation depicts information necessary for a given task (Kozma and Russell 2005).

In contrast to the fractions example, chemistry experts heavily rely on visual representations to solve chemistry tasks (Kozma and Russell 1997). Hence, the goal in using T+MV representations in chemistry instruction is not only to promote abstract understanding of the content, but also to promote students' learning of a "visual language" (Kozma and Russell 2005; Taber 2013). In addition, the chemistry education literature emphasizes the importance of becoming fluent in using individual visual representations (i.e., visual fluency) and in readily and effortlessly translating among multiple visual representations (i.e., connectational fluency; (Hinze et al. 2013a; Justi et al. 2009). In addition, meta-representational competencies are considered an important learning goal in and of itself: students need to learn which visual representation depicts which particular concepts, and when depicting these specific concepts is important for the given task (Gilbert 2005; Justi et al. 2009; Kozma and Russell 2005).

It is also important to note that some of visual representations in chemistry have significant information overlap, for example, ball-and-stick models and space-filling models depict atoms as spheres that correspond to their molecular volume, and both depict bond angles. They differ only with respect to how salient they make these two aspects: ball-and-stick models make it easier to see bond angles and bond types, whereas space-filling models make it easier to see the molecular volume. Because both are used in chemistry communities of practice, students typically encounter both ball-and-stick models and space-filling models during instruction.

A further goal in chemistry education is to understand the role of historical visual representations that are no longer used by experts. These visual representations are no longer in use because they have proven inaccurate (e.g., Bohr models that show electrons as orbiting a nucleus). Yet, they remain important because they had a significant impact on scientific discoveries in chemistry (e.g., on electrostatic forces) (Gabel and Bunce 1994; Nakiboglu 2003; Rocke 2010). Thus, chemistry is a domain in which representational competencies are a crucial aspect of domain expertise. Hence, the acquisition of representational competencies is a goal in and of itself.

Summary

This comparison illustrates that the role of particular representational competencies in education differs between the fractions and chemistry domains, and that these differences align with the socio-cultural practices of the use of visual representations in these domains. In comparison

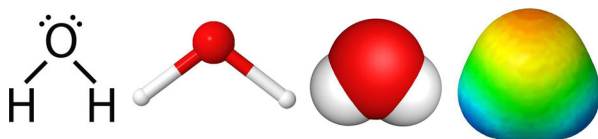


Fig. 2 Commonly used visual representations for bonding in chemistry: Lewis structure, ball-and-stick model, space-filling model, and electrostatic potential map of the water molecule (from left to right)

to fractions education, chemistry puts a stronger emphasis on the importance of visual fluency, connectional fluency, and meta-representational knowledge, because these representational competencies are a critical aspect of expert practices in chemistry.

In sum, this comparison provides indirect evidence for hypothesis 7, suggesting that T+MV scenarios should only be more effective at enculturating students into community practices than T+SV scenarios if the use of each visual representation aligns with its use in the target community of practice. For example, consider the T+MV scenario in Fig. 1c. Suppose that a student has acquired representational competencies for the pie chart and the rectangle but not for the numberline (hypotheses 1–5). Further, suppose that the numberline does not add informational value to the student's understanding of task 3 (hypothesis 6). Suppose, however, that the ability to place fractions on numberlines is an important aspect of mathematics expertise (e.g., when experts graph functions) (hypothesis 7). Consequently, the T+MV scenario may yet be more effective than the T+SV scenario because the student will acquire representational competencies relevant to the target community. However, a direct test of hypothesis 7 is still outstanding.

Discussion and Future Directions

The goal of this article was to investigate under which conditions T+MV may be more effective than T+SV scenarios in enhancing learning of content knowledge. This article approached this question from the perspective of the representation dilemma; that is, it investigated how students can learn *about* visual representations (i.e., acquiring representational competencies) while learning *from* visual representations (i.e., acquiring content knowledge). To this end, this article reviewed which representational competencies are involved in students' learning of content knowledge (Table 1). Two conceptual representational competencies acquired via verbally mediated sense-making processes were identified: visual understanding, connectional understanding. Two perceptual representational competencies acquired via nonverbal inductive learning processes were identified: visual understanding and visual fluency. Finally, meta-representational competencies build on both conceptual and perceptual competencies. Second, focusing on learning processes allowed analyzing how the number of visual representations affects the role these representational competencies play during students' learning of content knowledge. Specifically, contrasting T+SV and T+MV scenarios revealed that visual understanding is a facilitator of students' learning in T+SV scenarios, but a necessary prerequisite for learning in T+MV scenarios. Visual fluency plays a stronger role as a facilitator in T+MV scenarios. Connectional understanding and connectional fluency are facilitators of learning only in T+MV scenarios. Meta-representational competencies are more complex in T+MV than in T+SV scenarios and may therefore be a more important facilitator of students' learning in T+MV than in T+SV scenarios. This comparison yielded seven hypotheses that specify when T+MV scenarios might be more effective than T+SV scenarios. Empirical evidence for each of these hypotheses was reviewed (Table 2).

This review suggests several open questions that should be addressed in future research. First, it is worth noting that most support for these hypotheses comes from *indirect evidence*. Future experiments should empirically test these hypotheses in a variety of domains.

Furthermore, the hypotheses emerged from considerations about how conceptual and perceptual representational competencies interact with one another. However, we still know relatively little about these interactions. There are at least six important interactions that we

should investigate in future research. First, we need to investigate *interactions between sense-making processes and inductive learning processes* in order to provide effective instructional support that helps students benefit from T+MV representations. For example, if connectional fluency requires that students have connectional understanding, we may have to provide connectional-understanding support before connectional-fluency support. We know little about these interactions because research on sense-making of concepts and inductive learning of perceptual knowledge has been separate in the area of multiple external representations. This separation is somewhat surprising because there is a longstanding tradition in other areas of cognitive psychology to investigate relations between perceptual learning and conceptual learning (Goldstone and Barsalou 1998; Goldstone et al. 1997). In general, research on learning with visual representations would benefit from integrating these two perspectives. In particular, given the larger role of perceptual competencies for T+MV scenarios, integrating perceptual learning perspective into this research may yield interesting insights into differences between T+SV and T+MV scenarios.

Second, we need to investigate how processes of using *individual visual representations* interact with processes of *connection making among visual representations*. It was argued that visual understanding and fluency can promote connection making, while connectional understanding and fluency can also promote visual understanding and fluency. An open question is how much prior visual understanding is necessary so that a student can benefit from connection making tasks. Is a minimal level of visual understanding sufficient in enabling students to make connections? Or is a high level of visual fluency in at least one visual representation necessary, so that students can learn a second visual representation through connection making?

Third, it was argued that *meta-representational competencies* build on visual understanding, visual fluency, connectional understanding, and connectional fluency. However, it is an open question to what extent these representational competencies are required for the acquisition of meta-representational competencies, and whether teaching students about meta-representational competencies can enhance acquisition of other representational competencies as well. One important open question is how the student's prior representational competencies interact with their students' goals when acquiring meta-representational competencies. For example, it may matter whether the student prioritizes learning goals versus performance goals (Pintrich 2000). A student with performance goals may always choose the visual representation he/she is most fluent with because this choice will most likely lead to success in problem solving. In contrast, a student with learning goals may be more willing to choose the visual representation he/she is less fluent with because this choice has the highest pay-off in terms of learning gains. Should instructional support for meta-representational competencies take students' learning goals into account, and—if so—how? Another open question is whether students need connectional understanding before they can benefit from instructional support for meta-representational competencies. As mentioned above, support for meta-representational competence may enhance students' understanding of connections because it requires students to reflect on the differences between visual representations. However, comparing visual representations is cognitively demanding and may lead to cognitive overload unless students have at least some prior level of conceptual understanding. Yet, we still know little about what level of prior connectional understanding is necessary for students' benefit from support for meta-representational competencies.

Fourth, when we consider interactions among representational competencies, we need to take into account that they are *socially mediated and shaped by socio-cultural conventions*. As argued above, the relative importance of representational competencies is defined by the practices of the target community. From a socio-cultural perspective, the goal of instruction is to enable students to become part of scientific and professional communities, which implies that instructional interventions should reflect the cultural conventions and social functions the visual representations serve in the target community. More research is needed to investigate how practices in expert communities affect instructional support for learning with multiple visual representations.

For educational psychologists, incorporating a socio-cultural perspective would require that we understand how visual representations are used within the target community. This involves considerable effort: we have to delve deeply into discipline-based literatures that describe the educational goals with which visual representations are used. It would be helpful for the educational psychology research community to consolidate such discipline-based literatures into an overview across potential target domains. Such an overview would allow our research community to investigate in what ways socio-cultural aspects affect the generalizability of our findings about learning with multiple visual representations. Such an overview would also be useful to make predictions as to which types of instructional support for representational processes may be most effective for a given target domain. For example, based on the educational psychology literature, we might predict that students may not benefit from multiple visual representations if they have considerable information overlap, because the benefits (e.g., little additional information) do not outweigh the costs (e.g., effort necessary to make connections). Therefore, we may recommend using only one visual representation in instruction. Yet, if experts in the target community use different visual representations that have considerable information overlap, it is still necessary that students who are trained to become part of that community learn with these visual representations. In this case, we may recommend different types of instruction for different students. Those students who aim for a career in the discipline (e.g., biology majors) should learn with T+MV representations, and they should receive appropriate instructional supports (see section 5). Students who do not aim for a career in the given domain (e.g., nonscience majors in a science course) may benefit more from T+SV than T+MV representations. To date, it remains an open question how best to map requirements that result from discipline-specific practices to instructional interventions that are based on educational psychology research.

Fifth, it is still an open question to what extent representational competencies *transfer* across representations (NRC 2006). This review suggests that visual understanding and connectational understanding may transfer to visual representations that have similar feature-referent mappings (Gentner et al. 2003). Based on this argument, it was suggested that those competencies that result from sense-making processes (i.e., visual and connectational understanding, meta-representational knowledge) may transfer more easily than those that result from inductive processes (i.e., visual and connectational fluency), because sense-making processes involve principle-based reasoning and structure mapping, whereas inductive processes involve experience-based learning that is not verbally mediated. In contrast, visual fluency and connectational fluency may not transfer to novel visual representations (Ackerman 2003; Gibson 1969, 2000) or to other cultural contexts that use different types of discipline discourse (Carragher and Schliemann 2002). Yet, these theoretically motivated arguments remain to be tested empirically. The question of transfer is interesting also with respect to differences between STEM domains. In some domains, instructors may expect students to transfer

representational competencies across similar visual representations because there are certain conventions that are used across different types of representations (e.g., the ball-and-stick and space-filling models in Fig. 3 use the same color conventions to denote atom identity). Furthermore, the question of whether visual and connectional fluency transfer across topics, domains, or representation types, is important because it may affect an educator's decision to invest instructional time for representational competencies.

Finally, we need to consider potential *moderators* of students' acquisition of representational competencies. It is well established that spatial skills affect students' learning with visual representations and their ability to acquire representational competencies (NRC 2006; Stieff 2007; Uttal et al. 2013). Integrating information from visual representations into students' mental model of the content knowledge requires that students hold visual information in working memory and connect it to information in long-term memory (Hegarty and Waller 2005). Because spatial skills describe this very ability, cognitive load imposed by mental model integration is higher for students with low than with high spatial skills. Research on spatial skills has yielded instructional supports for spatial skills, but we do not yet know how spatial-skills supports interact with instructional supports for the representational competencies reviewed above.

Limitations

This review article has some limitations that should be noted. First, the review focused on visual representations that use *purely* perceptual devices as visual codes and that are encoded visually. Several types of representations do not meet these criteria because they use a *combination* of text-based and visual codes and are thus (at least in part) encoded verbally—such as maps, networks, matrices, or flow diagrams. We still know very little about how the use of such “hybrid” representations affects students' learning. Future research should investigate how students' understanding and fluency in using such representations affects their learning in STEM. In particular, hybrid representations often incorporate symbolic conventions that reoccur across multiple types of hybrid representations. Thus, students may have to understand and become fluent with symbolic conventions to process the visual aspects of hybrid representations. Another interesting aspect of hybrid representations relates to socio-cultural aspects. Some STEM domains rely on hybrid representations more so than others (e.g., flow diagrams are common in computer sciences). It would be interesting to map out the types of text-based, visual, and hybrid representations used in various domains because that would allow us to conduct research on how the use of hybrid representations affects students' use of visual representations and vice versa. To the extent that hybrid representations incorporate the same visual codes as some purely visual representations, understanding and fluency with hybrid representations may enhance students' understanding and fluency with visual representations, thereby enhancing how they benefit from multiple visual representations overall. One might also argue that understanding and fluency with visual representations may enhance students' benefit from hybrid representations.

Second, this article does not distinguish between *different types of text-based representations*. In most of the reviewed research, the text-based representation was written text. It may matter whether other types of text-based representations such as equations play a role during how students learn with visual representations. Future research should expand the space of scenarios covered in this article (see Fig. 1) to differentiate between single text-based and multiple text-based representations in addition to single and multiple visual representations.

A third related comment regards the focus on *visualization-specific competencies*, rather than general competencies such as spatial skills. For example, I consider the previously mentioned spatial skills as a general competency because they involve the ability to imagine and mentally transform any type of spatial information and to imagine any type of object from multiple perspectives in space (Hegarty and Waller 2005; NRC 2006). Although spatial skills play an important role in students' learning with visual representations and predict students' success in STEM (Hinze et al. 2013b; NRC 2006), they are not at the focus of this article because they are not tied to specific visual representations (e.g., understanding a pie chart versus understanding a numberline) but can transfer across a wide range of visual representations (Uttal et al. 2013). For an extensive review of research on spatial skills, refer to (Uttal et al. 2013).

Finally, this article assumes that most learners are *fluent readers*. International assessments such as PISA illustrate that this is a simplifying assumption that may not always hold (NCES 2011; OECD 2010). I am not aware of research that has investigated the effects of reading fluency on students' learning with visual representations, but this would be an interesting area for future research. For example, do students who are not fluent readers engage more willingly with visual representations? Can visual fluency help students acquire fluency in reading?

Conclusions

Visual representations can help students learn, but they can also confuse students because students often have to learn new content from new visual representations they do not yet fully understand (called the "representation dilemma"). To learn content knowledge *from* visual representations, students therefore need representational competencies: knowledge *about* how visual representations depict information about the content. Multimedia learning theory has focused on prior representational competencies that are prerequisites to students' learning from visual representations. This article extends multimedia learning research by integrating cognitive learning theories and socio-cultural theories to gain insights into the learning processes through which acquire representational competencies *while* they learn content knowledge. This review revealed a number of representational competencies that students acquire via socially mediated verbal *sense-making processes* and nonverbal *inductive learning processes*. The focus on these learning processes provided the basis for an analysis of how the presence of multiple versus single visual representations the role these competencies play during students' learning of content knowledge. This analysis yielded seven hypotheses that describe conditions under which we may expect multiple visual representations to be more effective than a single visual representation for students' learning of content knowledge. Finding that most evidence for these hypotheses is indirect, open questions about the conditions for the effectiveness of multiple visual representations were discussed. Considering these conditions in the design of learning activities will likely help instructors and students navigate the representation dilemma, help students learn from visual representations, and thereby help them succeed in STEM.

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